SUPERVISED MACHINE LEARNING

CUSTOMER CHURN PREDICTION

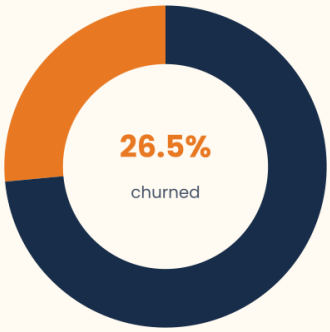
Step 1: Problem Framing & Theory Notes

Multi-Algorithm Classification Showdown

KNN, Naive Bayes, SVM and Decision Tree for predicting customer churn at a business like Jio or Airtel.

Churn vs stayed

share of customers in the Telco dataset



Why it matters

cost of a wrong prediction

False Negative — **HIGH cost**

False Positive — **low** — lose customer + CLV

26.5%
customers churned

73.5%
customers stayed

4
algorithms: KNN, Naive Bayes, SVM, Decision Tree

01 Frame
Churn, CAC vs CLV

02 Evaluate
Confusion matrix & imbalance

03 Choose
Algorithms & metrics

Q Step 1: Problem Framing & Theory Notes

1 What is Customer Churn? Why is predicting churn valuable for a business like Jio or Airtel? (Mention CAC vs CLV in your answer.)

Answer:

- Churn means a paying customer leaves the service (stops the subscription).
- Getting a new customer (CAC - Customer Acquisition Cost) is much more expensive than keeping an old one.
- Every churned customer also takes away his future revenue (CLV - Customer Lifetime Value).
- If Jio/Airtel can predict who will leave, the retention team can give offers in time and protect the revenue.

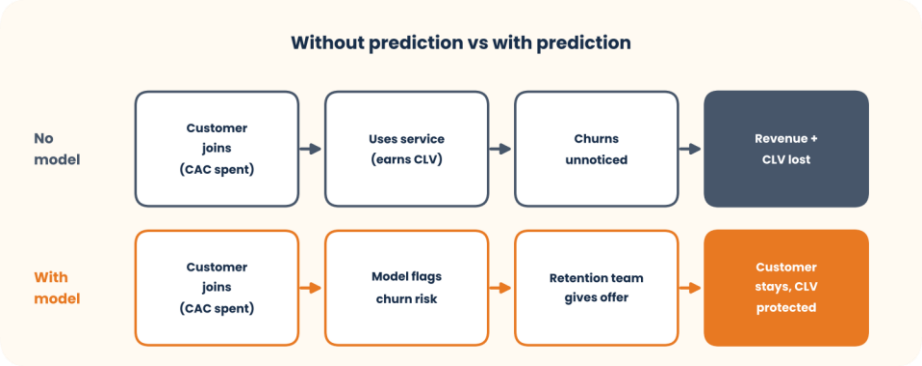


Figure 1: How churn prediction lets the retention team act before CAC and CLV are lost

2 Define the four cells of a Confusion Matrix (TP, TN, FP, FN). In the churn context — what is the business cost of a False Negative? What about a False Positive?

Answer:

- TP: customer actually churned and model predicted churn. TN: customer actually stayed and model predicted stay.
- FP: customer actually stayed but model predicted churn. FN: customer actually churned but model predicted stay.
- Cost of FN is high: we lose the customer and his CLV, and we never tried to save him.
- Cost of FP is low: we only waste a small offer or a call on a customer who would stay anyway.

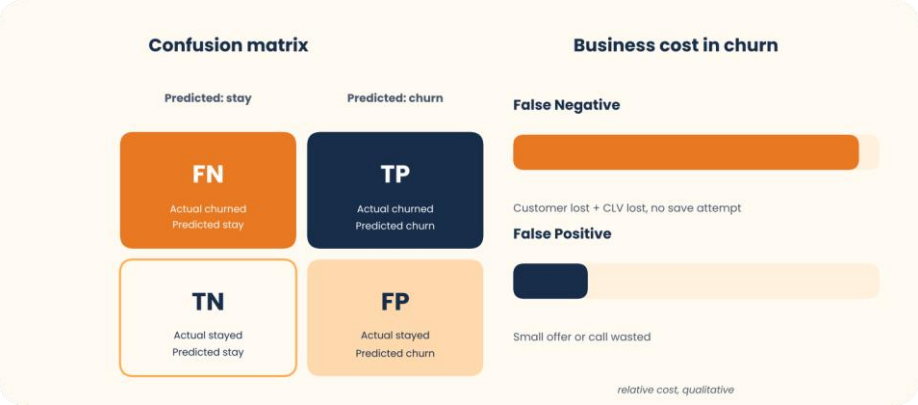


Figure 2: The four cells of the confusion matrix, with the relative business cost of FN and FP

3 What is Class Imbalance? Does the Telco Churn dataset have it? How does SMOTE address it?

Answer:

- Class imbalance means one class has far less data than the other class, so the model becomes biased to the majority class.
- Yes, Telco data is imbalanced - about 26.5% customers churned and 73.5% stayed (confirmed in Step 2).
- SMOTE creates new synthetic samples of the minority class using its nearest neighbours, so both classes become equal in training data.

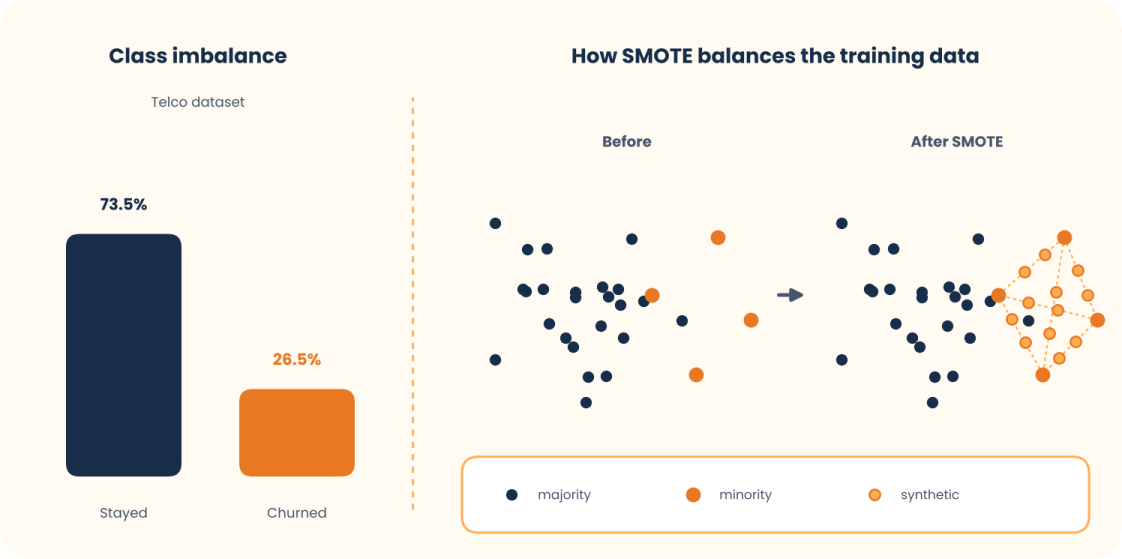


Figure 3: The 26.5% / 73.5% split in the Telco data, and how SMOTE adds synthetic minority samples

4 Briefly explain how each of the 4 algorithms you will use works — KNN, Naive Bayes, SVM, Decision Tree (2–3 lines each).

Answer:

- KNN: it stores all the training data. For a new customer it finds the k nearest customers and takes the majority class among them.
- Naive Bayes: it uses Bayes theorem to find the probability of each class given the features, and it assumes all features are independent.
- SVM: it finds the best boundary (hyperplane) with the maximum margin between the two classes. With the RBF kernel it can make a non-linear boundary.
- Decision Tree: it keeps splitting the data with yes/no questions on features (like Contract = Month-to-month?) and each leaf gives the final class.

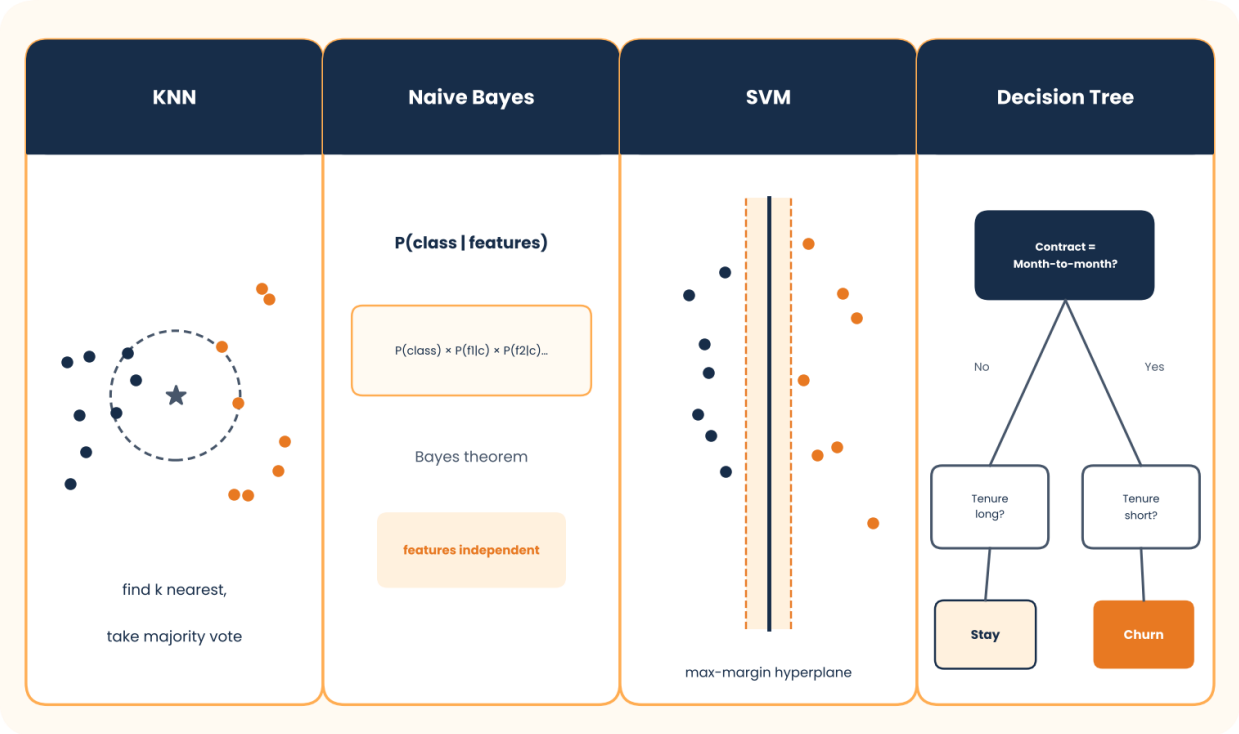


Figure 4: How KNN, Naive Bayes, SVM and a Decision Tree each separate churners from stayers

5 When would you choose Precision over Recall as your primary metric? Give a churn-specific answer.

Answer:

- Precision is more important when the retention action is costly or the calling capacity is limited.
- Example: if the team can call only ~500 customers a day or gives a big discount, wrongly flagged loyal customers waste money and call slots.

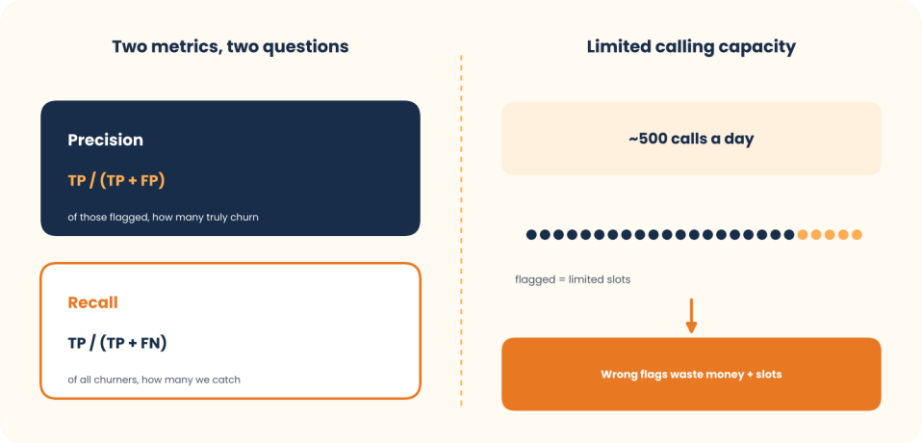


Figure 5: Precision and Recall, and why limited calling capacity favours Precision